Circumcision, Autonomy and Public Health

Brian D. Earp*, Yale University and The Hastings Center
Robert Darby, Independent Scholar

*Corresponding author: Brian D. Earp, Yale University, 2 Hillhouse Ave. New Haven, CT 06511, USA. E-mail: brian.earp@yale.edu.

Male circumcision—partial or total removal of the penile prepuce—has been proposed as a public health measure in Sub-Saharan Africa, based on the results of three randomized control trials (RCTs) showing a relative risk reduction of approximately 60 per cent for voluntary, adult male circumcision against female-to-male human immunodeficiency virus (HIV) transmission in that context. More recently, long-time advocates of infant male circumcision have argued that these findings justify involuntary circumcision of babies and children in dissimilar public health environments, such as the USA, Australasia and Europe. In this article, we take a close look at the necessary ethical and empirical steps that would be needed to bridge the gap between the African RCTs and responsible public health policy in developed countries. In the course of doing so, we discuss some of the main disagreements about the moral permissibility of performing a nontherapeutic surgery on a child to benefit potential future sexual partners of his. In this context, we raise concerns not only about weaknesses in the available evidence concerning such claims of benefit, but also about a child's moral interest in future autonomy and the preservation of his bodily integrity. We conclude that circumcision of minors in developed countries on public health grounds is much harder to justify than proponents of the surgery suggest.

Introduction

Male circumcision refers to the partial or total removal of the penile prepuce (foreskin) (Taylor *et al.*, 1996; Cold and Taylor, 1999). There is little disagreement among ethicists about the legitimacy of therapeutic circumcision—i.e. circumcision intended to address an extant pathology that has not responded to conservative treatment—nor is there much debate over the ethics of adult circumcision insofar as consent is fully informed (a condition that is not always met; Fox and Thomson, 2010; Sidler *et al.*, 2017). What does engender controversy are circumcisions performed on healthy babies or young boys who are not old enough to provide their own consent, most commonly for sociocultural or religions reasons rather than out of medical necessity (Davis, 2001, 2013; Benatar and Benatar, 2003; Fox and Thomson, 2005; DeLaet, 2009; Ben-Yami, 2013; Earp, 2013; Foddy, 2013; Mazor, 2013; Savulescu, 2013; Van Howe, 2013; Darby, 2015a; Jacobs and Arora, 2015; Ungar-Sargon, 2015; Svoboda *et al.*, 2016). In a recent paper, Azim McMath (2015a: 687) argues that while much of the controversy surrounding such nontherapeutic circumcision stems from disputes about empirical matters, some of it has to do with more theoretical questions concerning the autonomy of the child:

> while still an infant, the child cannot act autonomously. Nor has he ever been able to do so. However, if all goes well, he will eventually develop the capacity for autonomy. In addition, many of the professed benefits of circumcision, such as reduced risk of HIV infection, will occur primarily after the child becomes autonomous. Furthermore, there may be alternative means of achieving similar benefits, such as the responsible use of condoms, which depend largely on the child's future autonomous choices. In these circumstances, what does respect for autonomy require?

McMath argues that while a child's interest in future autonomy should generally be respected in relation to his own interests, the well-being of other parties may require that his autonomy be overridden in the interests of public health. At the same time, McMath seems conflicted about whether the seriousness of the threat of human immunodeficiency virus (HIV), especially in developed countries, can in fact justify the sacrifice of individual freedom that is entailed by circumcision in infancy or early childhood (that is, the freedom to make one's own decision about whether to undergo an elective genital surgery at an age of understanding; McMath,

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2015b). In this context, McMath's discussion about the child's interest in making decisions that reflect his mature preferences and values when he is older is compelling (McMath, 2015a: 689). But when considering arguments for paternalism in the name of public health, we suggest that McMath moves too quickly from certain empirical premises to associated policy proposals, skipping over gaps in evidence as well as important questions of value.

We do not mean to pick on McMath. In fact, his paper represents a good-faith attempt to navigate between the extreme and often polarized positions that have been articulated in the recent literature (for discussions, see Collier, 2012; Earp, 2015a). Even so, we claim, he adopts some of the same problematic assumptions about both the available data concerning health benefits and risks associated with circumcision, as well as their ethical implications, that have come to characterize the wider debate (Fox and Thomson, 2010). Using McMath's paper as a case study, then, our present aim is to slow the reel down to examine some of the intervening steps between data analysis and practical healthcare recommendations that must be taken into account.

Such an exercise, we hope, will have relevance not only for the debate over circumcision but also for other areas of health policy where there may be trade-offs between attempts at health promotion and respect for individuals' autonomy. In the course of our argument, we will touch on some of the main ongoing ethical disagreements concerning the permissibility of performing nontherapeutic infant or childhood male circumcisions, particularly in light of conceptual questions about the nature of a child's right to bodily integrity. First, however, we provide some historical background to frame the contemporary debates.

Background

Male circumcision originated in prehistory as a ritual practice (Glick, 2005). It was long regarded by its supporters as conferring primarily symbolic, spiritual or social benefits. During the Victorian period, however, doctors in Britain and the USA introduced circumcision not only as a measure of moral hygiene—dulling the sexual organ was thought to discourage childhood masturbation—but also as an intended preventive against certain physical diseases in adulthood (Darby, 2005). Routine circumcision reached a peak in popularity in the mid-20th century, but was subsequently rejected by the majority of child health authorities by the 1970s. Since that time, with the exception of the USA, it has largely been abandoned in the Anglophone countries that originally took it up (Wallerstein, 1985; Gollaher, 2000).

In the past few decades, facing increased scrutiny from legal theorists and medical ethicists as well as criticism from human rights advocates (Abu-Sahlieh, 1994; Denniston and Milos, 1997; Price, 1997; Boyle *et al.*, 2000; Cruz *et al.*, 2003; Sommerville, 2004), supporters of circumcision have sought to revive the 'medical benefits' narrative, casting the procedure as a secularly defensible measure of individual and public health, as opposed to solely a religious practice (Gollaher, 2000; Miller, 2002; Glick, 2005; Earp, 2017a). The search for benefits has been most fruitful in the USA, where, as noted, circumcision was adopted as a quasi-medicalized cultural norm in the late 19th century (Gollaher, 1994; Hodges, 1997). The acceptance and perpetuation of the procedure by key actors within the US medical community at that time and thereafter established the procedure as a ubiquitous birth custom, viewed as no more remarkable than clipping the umbilical cord after delivery. Consequently, the circumcised penis became the social norm (Hodges, 1997; Waldeck, 2003). There is now a vast literature, disproportionately generated by American doctors and researchers, purporting to show that circumcision is at least partially protective against a wide range of diseases and other problems. These range from urinary tract infections in early childhood, through sexually transmitted infections (STIs)—including HIV—after sexual debut, to penile cancer in old age (Wallerstein, 1980; Hodges, 1997; AAP, 2012).

Against this backdrop, a prominent feature of much current circumcision advocacy has been the underlying assumption that the biomedical evidence concerning such health benefits 'speaks for itself'. Once the data are available, that is, many advocates seem to assume that certain medical or public health policies should follow automatically as a logical consequence (Schoen, 2006; Tobian and Gray, 2011; for discussion, see Earp and Darby, 2015). This tendency has been particularly apparent in arguments that rely on evidence from three randomized controlled trials (RCTs) carried out in the early 2000s in Sub-Saharan Africa (Auvert *et al.*, 2005; Bailey *et al.*, 2007; Gray *et al.*, 2007). These RCTs provided evidence that voluntary circumcision of adult men may reduce the risk of female-to-male HIV transmission in areas with high rates of such heterosexual transmission and a low baseline prevalence of circumcision. As Aaron T. Norton argues (Norton, 2017: 3), the epidemiologists who conducted these trials, upon seeing the results, also saw a direct path to policy recommendations,

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their arguments reflecting 'an assemblage of scientific and ethical concerns'.

Stressing that circumcision had the potential to save lives, for example, the epidemiologists urged that to refuse to act was to act unethically. According to Nathan Hodson and Susan Bewley (2017), such alarmist rhetoric has featured prominently in many policy proposals surrounding HIV/AIDS, leading to a climate of 'AIDS exceptionalism' (i.e. the tendency to coopt the language of ethics to promote urgent action—and suppress dissent—without fully considering the moral dimensions of the problem or alternative ethical or policy views). Moreover, the epidemiologists seemed to believe the following:

> that science is the arbiter of ethical concerns via projections about future infections that rely not only upon modeling studies that employ the RCTs' statistical results, but also evidence of the foreskin's biomolecular susceptibility. If we know that foreskin is susceptible to infection by nature and that cutting it off saves lives, the implication is that not doing so kills, foregrounding the cause of death not in sexual behavior *per se*, nor in broader structural conditions, but in the body's own biological properties. (Norton, 2017: 3)

Faced with the RCT evidence, many public health officials, medical practitioners, journalists and even some bioethicists joined the rush to engage in extrapolation (Fox and Thomson, 2010). For example, if circumcision can lower the risk of so serious and, in some contexts, so prevalent a disease as HIV/AIDS, then, some seemed to think, it must be both medically desirable and ethically acceptable to perform it anywhere (Fox and Thomson, 2010; Bossio *et al.*, 2014, 2015). Moreover, if it 'works' for consenting adults, it should also be performed on infants and young boys (see Bossio *et al.*, 2015), and not only in response to individual (parental) demand, but through high-pressure public programs (Katisi and Daniel, 2015). In an interview (Norton, 2017: 3), one epidemiologist stated: 'A lot of men are calling it a mutilation. It's like, it *is*, but it saves lives, you know? There's a basis for it'. In this way, Norton remarks, 'concerns about the foreskin's loss pale in comparison to circumcision's ability to prevent death'.

In light of such thinking, mass circumcision as a strategy for HIV/AIDS control is now in full swing throughout the African continent, with the generous backing of American funders such as the Bill & Melinda Gates foundation (Garenne *et al.*, 2013; Bell, 2015; Giami *et al.*, 2015; Katisi and Daniel, 2015). Moreover, it has been construed as a straightforward instance of applying the discoveries of modern science to human betterment, akin to vaccination campaigns for polio or clean water for the prevention of cholera (Coghlan, 2009; Bill & Melinda Gates Foundation, 2017). But the relationship between evidence of health benefits associated with an intervention in a particular context, and the appropriate health policy to pursue on the basis of that evidence, whether in the original context or elsewhere, is not straightforward (Cartwright, 2007, 2011). As the philosophers of science Nancy Cartwright and Kathryn Joyce point out (Cartwright and Joyce, 2017: 3):

> studies don't produce policies. Rather, studies test hypotheses about whether an intervention will produce an effect [under certain conditions]. If the intervention yields a positive result—it produces the effect—in several well-conducted RCTs, we can say that the studies support the claim that an intervention works in the study settings. Now, we have a fact. But when will this fact count as evidence supporting use of an intervention?

Answering her own question in a separate paper (Cartwright, 2011: 1401), Cartwright writes that 'for policy and practice we do not need to know "it works somewhere." We need evidence for "it-will-work-for-us" claims: the treatment will produce the desired outcome in our situation as implemented here'. For this, in turn, we need 'a large and varied inductive base . . . lots of RCTs from different populations—plus reason to believe the observations are projectable, plus an account of the range across which they project'. In other words, while *efficacy* might be demonstrated under ideal or research conditions, there is no guarantee that an intervention will be *effective* in a real-world setting. Thus effectiveness trials, also known as translational research, are typically undertaken before generalized policy positions are taken. No such translational research was performed in regards to male circumcision and HIV infection.

Some advocates of circumcision seem to appreciate the need for a basis for projection. That is, they recognize that even if the RCT findings are taken at face value, they are not sufficient to show that circumcision will be effective—much less prudent on public health or ethical grounds—in other settings apart from Sub-Saharan Africa. This realization has led to 'attempts to generalize the results of the circumcision RCTs to novel contexts via appeals to the foreskin's biomolecular susceptibility' (Norton, 2017: 4). In other words, circumcision advocates have begun to argue that simply having a foreskin should be 'effectively considered a pre-disease state defined in relation to HIV risk' (Norton, 2017: 5).

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This conceptual shift represents a radical departure from how risk is normally understood. Unlike the identification of pre-cancerous cells in the cervix, say, that may be associated with an increase in the future risk of cervical cancer, 'arguments that the foreskin poses a risk do not rely on claims that the foreskin's cells have *become* compromised, but rather, on claims that the foreskin's "normal" cellular state is *already* risky' (Norton, 2017, pp. 5–6). Construing a healthy body part as in-and-of-itself a source of risk has proven strategically valuable for circumcision advocates: 'if HIV susceptibility is an inherent feature of the foreskin, it becomes easier to dismiss contextual factors that may have shaped the RCT results, and thus generalize not only from trial context to the real world, but across real-world contexts' (*ibid.*)

It is true that human genital mucosa, whether found on the bodies of males or females, may be penetrated by HIV (Cold and Taylor, 1999; Taube *et al.*, 2007; Hladik and Hope, 2009). It is not implausible, therefore, that reducing the surface area of such mucosa—in males, by excising the foreskin—would lower a person's risk of becoming infected with HIV by some amount, just as excising any other body part would reduce the risk of some disease affecting it, or introduced to other body sites through it (Earp *et al.*, 2017a; Kluge, 1994). But to slide from such a biological assertion to the proposition that circumcision is therefore on balance desirable as a personal or public health intervention is a non-sequitur (Darby, 2016). There are many possible responses to such claims, and the appropriate response will be determined not by science alone, but also by the relevant values of the individuals and societies in question (Douglas, 2009).

The very notion of what constitutes optimal health, for example, is subject to cultural variation (Levesque and Li, 2014), and a preference for (say) a modest reduction in the absolute risk of a treatable infection over certain kinds of sexual sensation or experiences of bodily integrity is also a culturally influenced priority (Dekkers *et al.*, 2005; Dekkers, 2009; Earp, 2017b, c; Earp and Darby, 2017). It is telling that in countries where circumcision is the norm, such as Saudi Arabia or the USA, health professionals tend to commend circumcision as a prophylactic against HIV (AAP, 2012; Alkhenizan and Elabd, 2009; see also Earp, 2015b), whereas in those countries and regions where surgically unmodified male and female genitalia are the norm—such as most of Europe, Britain, Canada, Brazil, Australia and New Zealand—health professionals do not regard circumcision as favorably and have by and large rejected it as an HIV control tactic (BMA, 2004; KNMG, 2010; RACP, 2010; Frisch *et al.*, 2013; CPS, 2015; Earp and Shaw, 2017).

Public health crises, such as the syphilis scare in 19th-century Britain no less than HIV/AIDS in contemporary Africa, have historically given rise to calls for drastic action in which the autonomy interests of individuals become subordinated to the greater good, with an attendant rise in the level of state or social paternalism (Darby, 2015b; Gonsalves and Staley, 2014; Hodges, 1997). There are certainly situations in which the demands of public health will require the restriction of individual autonomy—for example, quarantine in the case of highly infectious diseases; and even then governments must be on guard against overreaction (Cetron and Landwirth, 2005; Drazen *et al.*, 2014)—but it does not necessarily follow that the HIV/AIDS crisis in Sub-Saharan Africa is an instance of this proposition, nor that it warrants the heavy-handed promotion (Katisi and Daniel, 2015; Rudrum *et al.*, 2017) of an irreversible genital alteration whose risks have not adequately been studied (AAP, 2012; Bossio *et al.*, 2014; Earp, 2015c; Frisch and Simonsen, 2015a, 2016). Nevertheless, respect for autonomy entails that adult men should be allowed to elect circumcision for themselves, whether as an attempted form of prophylaxis against HIV or for any other reason. Problematically, however, as adult male volunteers have failed to materialize in sufficient numbers to meet the quotas set by circumcision backers, advocates are now pressing for the procedure to be performed on infants and young children, who cannot decline (Mavhu *et al.*, 2012; Hatzold *et al.*, 2014; Bailey *et al.*, 2017; Sidler *et al.*, 2017).

Some researchers have begun to resist what they perceive as an ill-advised circumcision policy juggernaut. For example, Kenneth Rochel de Camargo and colleagues have recently argued that the current approach to policy is both technocratic and authoritarian, running roughshod over the complexities of culturally sensitive and ethically responsible medical decision-making (de Camargo *et al.*, 2013; de Camargo *et al.*, 2015a, b; Giami *et al.*, 2015). Inadequate attention is given to social, behavioral and other contextual factors that are relevant to disease prevention and management (Van Howe and Storms, 2011; Giami and Perrey, 2012; Norton, 2013, 2017; Darby, 2015b). Bioethical and human rights principles are dismissed as nebulous (Fox and Thomson, 2010). And 'subjective' issues such as individual preferences and values regarding bodily aesthetics and genital integrity are deemed to be irrelevant (Darby and Cox, 2009; Richters, 2009; Earp and Darby, 2017).[1]

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Circumcision is not, however, 'just another' public health intervention, based primarily on evidence of net benefit rather than harm. As Peter Aggleton observes, circumcision 'has its roots in the deep structure of society. Far from being a simple technical act, even when performed in medical settings, it is a practice which carries with it a whole host of social meanings' (Aggleton, 2007: 15). These meanings may relate to rites of passage, religious customs, or questionable beliefs about hygiene, and are often 'a potent indicator of hierarchy and social difference'. Circumcision, then, is not a 'value neutral' act, but 'nearly always a strongly political act, enacted upon others by those with power'. It is for this reason that close ethical scrutiny of proposals by 'those with power' to enforce circumcision on the bodies of those without it is required.

A Right to Bodily Integrity?

To set the stage for a potential trade-off between child autonomy and public health, McMath raises the issue of a 'right to bodily integrity'. Critics of nontherapeutic circumcision sometimes claim that the procedure violates a child's right to bodily integrity (Androus, 2004; Denniston *et al.*, 2007; Ungar-Sargon, 2015), while supporters deny this (Benatar and Benatar, 2003; Mazor, 2013). Resolving this dispute is important because if nontherapeutic circumcision does violate a child's rights—whether to bodily integrity or anything else—its use as a public health measure would be much more difficult to justify.

There is disagreement in the bioethical literature about whether children have an absolute right to—as opposed to, say, a very strong interest in—having their bodily integrity preserved (Ludbrook, 1995; Mazor, 2013; Ungar-Sargon, 2015). But even if one simply grants that children do have such a right, it would not entail that one must never break the skin or otherwise alter the body of a child. As can be seen with so-called 'medically necessary' surgeries (such as might be needed to correct a heart defect, or even a therapeutic circumcision in the event of severe frostbite or incurable balanitis xerotica obliterans), it is usually permissible to infringe upon the bodily integrity of the child if doing so is clearly in the child's best interests (Kopelman, 2007; Maslen *et al.*, 2014).

What counts as being in the child's best interests, however, is a matter of dispute (Kopelman and Kopelman, 2007; Kopelman, 2014; Earp and Shaw, 2017). McMath raises the example of vaccination. As he notes, this is an intervention that breaks the child's skin without being medically necessary, and yet is widely considered to be consistent with the child's best interests (and certainly not a rights violation). Insofar as this observation is meant to pave the way for the acceptability of circumcision, however, the analogy is strained (Lyons, 2013). Those who argue for a 'right to bodily integrity' do not typically hold so sensitive a notion of bodily integrity in mind that they would regard it as being impermissibly infringed by a mere pinprick on the arm (especially when the prick is needed to deliver medicines that are expected to boost the child's immunity to disease while she or he is still in childhood). Rather, they have in mind a more robust sense of bodily integrity: the absence of substantial changes to the body that may reasonably be regarded as a harm (Jonsen, 1978; Earp and Darby, 2017; Earp, 2017c).

Parents who alter their children's bodies—as with nontherapeutic genital surgeries—typically do not intend to harm them. Rather, in most cases, they view the alteration as an enhancement (i.e. something that is expected to improve the child in some way, whether physically, spiritually or socially; Svoboda, 2013; Earp, 2016c). But many children—female, male, and intersex—whose genitals were altered without a strict medical need, grow up to regard the intervention or the associated bodily change as a diminishment or even a mutilation (Hammond, 1999; Lightfoot-Klein *et al.*, 2000; Bossio and Pukall, 2017; Earp and Steinfeld, 2017; Hammond and Carmack, 2017). This phenomenon has inspired a worldwide 'genital autonomy' movement that is largely fueled by such resentful individuals (Silverman, 2004; DeLaet, 2012; Wisdom, 2012). Such a polar 'flip' from intended enhancement to experienced net harm or mutilation appears to occur in a minority of cases across sexes (albeit with considerable variance depending on the context), especially when the sociocultural norms upholding such practices are relatively widespread and not typically questioned (Dreger, 1999; Gage and Van Rossem, 2006; Karkazis, 2008; Reis, 2009; Sircili *et al.*, 2010; Abdulcadir *et al.*, 2012; Shweder, 2013, 2016; Earp *et al.*, 2017b). Nevertheless, such extreme negative feelings appear to be more common as a response to nontherapeutic early childhood genital modifications than to other commonly cited forms of intended pediatric enhancement—such as cosmetic orthodontia—that also involve making permanent physical changes (Bossio and Pukall, 2017; Earp, 2015d, e; Hammond and Carmack, 2017; Mollov *et al.*, 2010). Why might this be so?

One possible explanation is that alterations that by necessity damage or remove sensitive genital tissue—or more generally, tissue that it is reasonable to regard as having value in and of itself (Kipnis and Diamond, 1998)—are more likely later to be perceived as harmful

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compared to other bodily alterations or intended enhancements that do not have such a necessary effect. In simplest terms, if the guaranteed or intended outcome of the procedure is to excise tissue that one might sensibly wish to have retained (see Earp and Darby, 2017), then the procedure is at a greater risk of being seen as 'mutilating' than one whose threat to such tissue is relatively minor or unintended. Thus, while cosmetic orthodontics do certainly alter the body for 'nonmedical' reasons—and while there are nontrivial risks associated with such alteration (e.g. infections, bleeding, poor cosmetic outcome)—the potentially adverse physical changes are either temporary or accidental to the procedure; they are not its intended effect (Alani *et al.*, 2015). Accordingly, there is no groundswell of resistance to aesthetic braces. More broadly, when the status of an intervention as an enhancement—as opposed to something that diminishes or causes harm—is stable across time and place and highly secure against possible changes in perspective, it is more likely to be in the child's best interests overall (Maslen *et al.*, 2014; Earp, 2016c).

It is the existence of this implicit, 'robust' sense of bodily integrity that is perhaps the main reason why piercing of girls' ears for cosmetic or cultural reasons does not arouse particularly heated moral arguments (Holm, 2004; Earp, 2012). Although such an intervention is nontherapeutic and does technically infringe upon the girl's bodily integrity, like cosmetic braces, it is normally considered too minimal to deserve special attention. Although there is momentary pain as well as some risk of infection, the 'final result' is a small hole that may very well close up if the child later decides she would like her earlobes to be hole-free. By contrast, male circumcision irreversibly removes between 1/3 and 1/2 of the motile skin system of the penis, consisting of highly innervated, touch-sensitive, functional and protective tissue (Taylor *et al.*, 1996; Werker *et al.*, 1998; Scott, 1999; Sorrells *et al.*, 2007; Kigozi *et al.*, 2009; Bossio *et al.*, 2016; Earp, 2016d). We do not wish to defend the practice of infant ear-piercing; but in terms of degree of invasiveness and permanence, ear piercings are far removed from a surgical operation that excises a substantial amount of erogenous tissue from a psychosexually significant external organ. The latter, compared to the former, is much more easily classified as a 'substantial change to the body that may reasonably be regarded as a harm' (see Box 1).

While these considerations do not vitiate McMath's point that it is the best interests standard rather than an overriding right to bodily integrity that is the relevant moral concern (for further discussion, see Carmack *et al.*, 2016), they do suggest that the *burden of proof* that circumcision really is in the child's best interests is considerably higher than the burden of proof that immunization is in the child's best interests (Lyons, 2013; Adler, 2016). Among other differences, vaccination does not (i) remove erogenous or any other healthy tissue; (ii) eliminate any bodily functions (such as the protective function of the foreskin and any and all sexual functions that involve manipulation of the foreskin); or (iii) involve the permanent surgical

Box 1. A brief overview of the penile prepuce (foreskin). Adapted from Earp *et al.* (2017b)

According to Cold and Taylor (1999: 34) the prepuce 'is an integral, normal part of the external [male] genitalia that forms the anatomical covering of the glans penis'. In this respect, it is similar to the clitoral prepuce in females, which similarly covers and protects the glans clitoris. One of its functions is to internalise the head of the penis including the urethral meatus, 'thus decreasing external irritation or contamination' (*Ibid.*). This feature may help to explain why meatal stenosis and other urethral stricture diseases, thought to be caused by abrasion of the exposed meatus, are far less common in genitally intact boys compared to circumcised boys (Frisch and Simonsen, 2016). Similar to the eyelids, lips, anus, and female genital labia, the penile prepuce is a 'specialized, junctional mucocutaneous tissue which marks the boundary between mucosa and skin . . . the unique innervation of the prepuce establishes its function as an erogenous tissue' (*Ibid.*).

In the adult organ, the foreskin has an average surface area of approximately 30–50 square centimeters (Kigozi *et al.*, 2009; Werker, *et al.*, 1998), constituting roughly half of the moveable skin system of the penis (Taylor *et al.*, 1996). Recent research using objective measures suggests that the foreskin is the most sensitive part of the penis to light touch, while also being significantly more sensitive than the head of the penis to sensations of warmth (Bossio *et al.*, 2016; Earp, 2016d). As it is an elastic, retractable sleeve of tissue, the foreskin can be manipulated during sex and foreplay, whether manually or orally, thus providing specific subjective sensations that some men regard as being highly pleasurable (Ball, 2006). Although 'the amount of genital tissue removed is variable . . . the penile prepuce is removed in nearly all male circumcisions' (Cold and Taylor, 1999: 34), thereby precluding such particular sensations.

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modification of a body part whose altered state is not uncommonly perceived as a diminishment rather than an improvement.

Paternalism and Harm to Others

Having established that the focus should be on what is in the child's best interests, rather than on broad or unqualified appeals to potential rights violations, McMath states that male children have an interest in reducing their risk of contracting HIV (McMath, 2015a: 687). Undoubtedly, they do. But this does not mean that they have an interest in being circumcised *in infancy* on such grounds, since there is no controlled evidence that infant circumcision—as opposed to adult circumcision—can in fact reduce transmission of HIV, much less to a meaningful degree outside of the context of Sub-Saharan Africa, where absolute rates of HIV transmission are among the highest in the world (Sidler *et al.*, 2008; Darby and Van Howe, 2011; Bossio *et al.*, 2014, 2015; Frisch and Earp, 2016). Since one's susceptibility to sexually transmitted infections is far more strongly governed by socio-behavioral mechanisms than by strictly biological mechanisms (such as the presence or absence of a foreskin), it cannot be assumed that a partial protective effect accruing from adult circumcision in Sub-Saharan Africa would also follow from circumcision of infants in other contexts (RACP, 2010).

But what if it could be assumed? In other words, what if, one day, there is convincing evidence that specifically infant circumcision in some setting can reduce the risk of female-to-male transmission of HIV in that setting? Would that justify infant circumcision? No, it would not. At best it would count in favor of such circumcision—it could be added to the benefit column—but the *weight* to be assigned to this benefit would depend on many factors, both ethical and epidemiological. For example, what is the absolute risk of contracting HIV through heterosexual intercourse in the relevant context? If it is very low, as it is in most contexts outside of Sub-Saharan Africa, then the weight to be assigned in this regard should also be low. To see why this is the case, consider the following analysis of HIV transmission in the USA by Sarah Bundick (2009):

> Heterosexual contact is estimated to be responsible for only 5, 250 new infections in men each year . . . there is no strong evidence that circumcision reduces the risk of male-to-female or male-to-male transmission via sexual contact [and it] is not related to HIV transmission by injecting drug use. Therefore, the current data suggest that the 5, 250 female-to-male transmissions are the only ones [even eligible] to be prevented by male circumcision. If we assume that all 5, 250 men who get HIV from a female sexual partner are not circumcised (though this is certainly not the case), the data suggest that about half of these infections—around 2, 625 infections or ∼5 per cent of new infections—may have been prevented if the men had been circumcised. If we then factor in the number of men who are circumcised when they are infected (approximately 70–80 per cent of American men are already circumcised), the number of infections that could have been prevented by circumcision drops considerably. Taken together, the data suggest that the number of HIV infections that could be prevented in the U.S. by promoting infant male circumcision is likely to be only in the hundreds per year—a tiny fraction of the estimated 50, 000 new HIV infections.

Such 'tiny' numbers are based on generous assumptions about the potential effectiveness of infant circumcision as an HIV preventive in the USA because the applicability of the African RCT data pertaining to adult males is simply being stipulated. Nevertheless, they demonstrate the importance of taking context—including baseline risk figures—into consideration when trying to assess whether (male) children have any interest at all in being circumcised as prophylaxis against HIV. But let us just assume that they do—that *some* weight should be assigned to this potential benefit of circumcision, however small. This still does not tell us whether the surgery is in the overall best interests of the child because additional weights would have to be assigned, not only to other purported benefits, but also to risks and harms, many of which have not been accurately estimated (and some of which, such as feelings of loss or resentment, are subjective in nature and therefore impossible to quantify; see Frisch and Earp, 2016).

Finally, the attitudes and values of the child himself with respect to risk-type, risk-comparison and alternative risk-reduction strategies also have to be factored in. For example, is a slight reduction in the absolute risk of contracting a rare and treatable infection that can also be avoided nonsurgically 'worth' the risk of a similarly rare surgical error resulting in damage to the head of the penis? Is it 'worth' the risk of removing too much foreskin tissue, leading to painful erections or penile curvature later in life? There is no objective answer to such questions. Instead, only the child himself when he is grown can assess such trade-offs, at which point he can take into consideration the fullness of his circumstances as well as his own values (Earp and Darby, 2017). What this suggests is that no amount of new evidence

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concerning either the benefits or risks of circumcision will be sufficient to determine the question of interests. There will always be a personal, subjective aspect to such judgments, which even McMath sees as counting in favor of refraining from elective circumcisions before an age of consent (McMath, 2015a: 689). He asks the reader to:

> [c]onsider the extent to which STI prevention constitutes a benefit. If the child grows up to be celibate, it will not be a benefit at all; if he grows up to be sexually profligate, it may be a significant benefit. The child himself, once he reaches an age of competence, will be in a better position to judge his own interests than his parents will be while he is still an infant.

Moreover, giving the child the opportunity to decide for himself at a later age can help resolve disagreements over values. As McMath notes, reasonable people disagree over what constitutes a harm or benefit. For example, 'Some people believe circumcision benefits the child by bringing him closer to God, while others disagree. In light of such disagreement, some commentators conclude that the parents should decide' (p. 689). But this does not follow:

> After all, the child will have an interest in living according to his own values, which may not reflect those of his parents . . . only the child himself, when he is older, can be certain of his values. Thus, if disagreement over values constitutes a reason to let the parents decide, it constitutes an even stronger reason to postpone the decision until the child himself can decide. (McMath, 2015a: 689)

Given these and other considerations, McMath concludes that we should be 'idealists' about the infant child's future autonomous choices insofar as they affect himself. But what if they affect others? This is where McMath brings in his argument about public health. Given the lack of any controlled evidence for a protective effect of infant circumcision against the acquisition of HIV, we can assume that McMath takes for granted something like the 'biomolecular' view of foreskin risk proposed by circumcision advocates (Norton, 2017). That is, based on the unproven assumption that it is the sheer lack of a foreskin in the circumcised men that was chiefly responsible for the RCT findings—and that this lack *per se* will result in a similar risk-reduction in any epidemiological environment—McMath argues that we must consider the interests not only of the child himself, but also 'anyone who, in future, has sex with him or shares needles with him' (McMath, 2015a: 687). We need to do this because, according to McMath, 'reducing one person's risk of contracting HIV—or any other infectious disease—also reduces the risk to many other people. Thus, we need to consider the impact of circumcision not only on the interests of the child, but also on public health'.

Now, by 'public health', McMath really appears to mean 'HIV prevention', since he does define the term and his examples almost exclusively concern the latter. But there are several issues being obfuscated here. First, simply having been circumcised, whether in infancy or later in life, does not entail that a man's risk of contracting HIV is lower than it otherwise would be, much less that he is less likely to transmit HIV to others. Part of the reason for this is that some circumcised men engage in 'risk compensation' (Riess *et al.*, 2010; Grund and Hennink, 2012; Feldblum *et al.*, 2015; Jung *et al.*, 2016; Kibira *et al.*, 2017; but see Shi *et al.*, 2017). This occurs when a circumcised man, presumably because he believes that his risk of contracting HIV is lower due to his lack of foreskin, comes to rely less on more surefire forms of safe sex, such as condom use or a reduction in the number of concurrent sexual partners. If he does reduce his reliance on these measures, he may actually increase both his and his partners' risk of infection, over and above whatever advantage he may initially have had (in this respect) in being circumcised. Moreover, the only RCT looking specifically at male-to-female transmission of HIV as a function of the male's circumcision status had to be stopped early due to futility: more female partners of circumcised men, compared to genitally intact men, were becoming infected with HIV, possibly due to early resumption of sex during wound-healing (Wawer *et al.*, 2009). If public health or harm-to-others is the concern, then, one needs to take such considerations into account (Dushoff *et al.*, 2011).

Failing to raise such caveats, McMath nevertheless maintains that the prospect of harm-to-others may play a legitimate role in justifying infant circumcision on public health grounds, presumably in any setting (since he does not specify an epidemiological environment). He is aware, however, that this line of thought could lead to policy conclusions that might be seen as overly paternalistic. To address this concern, he writes:

> People sometimes make autonomous decisions that result in risks and harms to others. There is nothing paternalistic about coercive intervention to prevent these other-regarding risks and harms. While people should generally be at liberty to assume risks to themselves, they should not generally be at liberty to impose risks on others. To the extent that the child's future choices may

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> harm others, and to the extent that circumcision may help to mitigate these harms, the procedure may be justified as a matter of public health. (McMath, 2015a: 689)

It is true that people are not generally at liberty to impose (undue or unwarranted) risks of harm on others. But the mere possession of an intact penis does not impose risk on anybody, and certainly not undue or unwarranted risk. In other words, McMath slides invisibly in this passage from harm caused by potential 'future choices' (such as failure to wear a condom), to harms caused by *simply having an unaltered penis*—i.e. the only factor that could be 'mitigated' by a circumcision. But as Norton notes, 'the presence or absence of a body part cannot reasonably be said to be the cause of HIV infection, but rather plays an indirect or "moderating" effect in the likelihood of infection that may or may not apply depending upon other conditions' (Norton, 2013: 276).

What is the more general implication of McMath's suggestion? In another paper, Norton writes: 'Policies aimed at making routine neonatal circumcision more accessible prioritize removing the "vulnerable" foreskin over what kinds of sexual practices or HIV prevention strategies [such as condom use] an infant may eventually engage in as a sexually mature person' (Norton, 2017: 7). In response to such observations, McMath notes that 'many people do not use condoms responsibly. Consider the fact that nearly half of all pregnancies in the developed world are unintended. We cannot necessarily assume that the child will autonomously decide to use condoms responsibly when he grows up' (McMath, 2015a: 688).

The reference McMath gives for unintended pregnancies is to a study that made no breakdown on the basis of whether or not the male was circumcised; it may well be that unintended pregnancies are more common among the circumcised. In fact, some studies suggest that circumcised men, compared to genitally intact men, are less likely to use condoms as well as more likely to experience condom slippage when they do (Richters *et al.*, 1995; Crosby and Charnigo, 2013; Frisch, 2016). Thus, although McMath argues that it is necessary to be a realist rather than an idealist in such matters, it is not obvious that the realist response requires us to accept the propriety of infant genital surgery, even on other-affecting grounds.

Instead, it leads us to consider how adult sexual partners might choose to negotiate the risks to which they may expose themselves: except in the case of rape, nobody is forced to have sex with a 'riskier' sexual partner.[2] Moreover, people do not have sex as embodied statistical averages: individual differences in sexual context and behavior are far more predictive of one's transmission risk than circumcision status, such that men's prospective partners would do well to focus on those other differences rather than on whether the men have had their foreskins removed (which, if they have, could create a false sense of security). To summarize, it is only consensual adult behavior—engaged in by both or all parties who may be affected—that presents a risk of harm. Sexually transmitted infections are not like sneezes: they cannot be spread throughout a school or a shopping center to unsuspecting third parties.

Tying these strands together, as Norton (2017: 15) notes, 'promoting circumcision to reduce risk for a given infant'—much less a possible future sexual partner of his—'is based on a number of assumptions at the moment the decision is made, assumptions that reflect the emergence of circumcision status as an HIV risk category rooted not in sexual identity or behavior but in the body's own biological properties'.

Yet such assumptions about risk lead to absurd consequences. Any healthy part of person's body could, in principle, be 'misused' to inflict harm on other people. For example, one's finger could be used for poking out someone's eye. But since it can also be used for morally unproblematic purposes, and it serves many important functions, infant finger removal should not be entertained. In other words, it is problematic to remove an unoffending part of a person's body *just in case* he might use it in a way that would increase the risk of harm to others, especially insofar as those other people also possess agency in protecting themselves from any such harm.

When looking at an infant, we do not know what his 'future choices' regarding sex and sexual behavior will be. If, as McMath suggests, it is nevertheless permissible to excise his healthy foreskin without his consent, on the assumption that its sheer retention on his body puts others at risk of harm, then why is it not permissible to sever an adult man's foreskin, whose future sexual behavior is much easier to predict? If the mere possession of intact genitalia is meant to be understood as 'risky' and potentially other-harming, then—on public health grounds—there is equal, if not more, reason to engage in coerced circumcision of men.

But this would be criminal assault (Merkel and Putzke, 2013). If it is not permissible to coercively circumcise a man on public health grounds, it is questionable why we should be able to circumcise an infant or small child on the same grounds, simply because the latter is pre-autonomous and can neither give nor

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withhold his consent. In other words, given that healthy genital tissue is *prima facie* valuable, and that the mere possession of intact body parts cannot reasonably be construed as presenting a special kind of risk, a stance of skepticism toward proposals such as McMath's seems in order.[3]

McMath himself, writing subsequently on the *Journal of Medical Ethics Blog*, does appear to concede some of the problematic implications of his view. Responding to the release of a non-peer-reviewed draft policy by the US Centers for Disease Control and Prevention (CDC) stating that the benefits of circumcision outweigh the risks (CDC, 2014), McMath (2015b) writes:

> the CDC relies far too heavily on HIV prevention as a justification for infant circumcision. In the United States, the adult prevalence of HIV is 0.6 percent – not very high by global standards. Furthermore, as the CDC itself admits, only 10 percent of new infections are transmitted sexually from a female to a male . . . as I have argued, most of these initial infections cannot justify infant circumcision; it is the extent to which these infections increase risks to others that is ethically relevant. All of this suggests it would take very many (perhaps thousands) of circumcisions to prevent one new ethically relevant HIV infection in the United States. The public health justification for infant circumcision, at least in terms of HIV prevention, seems much weaker than the CDC's enthusiasm would suggest.

Conclusion

McMath's basic argument—that individual rights and interests must sometimes give way to the general good—is in principle sound, but the question is whether it is correctly applied to the instance of child circumcision. Nobody but an extreme libertarian would disagree that it can be reasonable to restrict an individual's freedom of action—including with respect to certain decisions about the state of his own body—if the general welfare interests are substantial enough and the burdens placed on the individual are proportionate to the gravity of the problem and equitably distributed.

In the public health field, a good example is quarantine—once a regular feature of trans-oceanic travel, and still seen sometimes today in regions of severe infectious epidemics. Under quarantine an individual's liberty is restricted for a short period to protect other members of society from diseases that he or she is or may be carrying. Since all individuals have an interest in freedom of movement, society or other authorities (parents in the case of children) must have good reason to override this interest. But unlike circumcision, quarantine is short-term and has no lasting consequences (provided proper access to medical care and other necessities); although it is a harm, it is a minor harm that imposes a short-lived burden. Once the prescribed period has elapsed, an individual can resume his life at the point where he left off; the only lasting effects are memories. It is also equitable in that all persons at risk are subject to the same restraint. Once the issue is cast in these terms it is clear that the public health justification for quarantine is not applicable to childhood male circumcision.

The inadequacy of the public health argument for circumcision of boys was well summarized in a paper as early as 2002. In 'Prophylactic Interventions on Children: Balancing Human Rights with Public Health', Hodges *et al.* performed a systematic analysis of the rights and interests of children in relation to public health benefit (Hodges *et al.*, 2002). With a view to laying down a set of standards for how conflicts between the demands of public health and the principles of human rights might be resolved, and observing that prophylactic interventions on children were traditionally justified on the grounds of 'best interests of the child' and/or 'public health', Hodges *et al.* proposed two sets of criteria that should be met for an intervention along these lines to be acceptable.

The criteria for the 'best interests of the child' argument were (i) presence of clinically verifiable disease, deformity or injury; (ii) least invasive and most conservative treatment option; (iii) net benefit to the patient and minimal negative impact on patient's health; (iv) competence to consent to the procedure; (v) standard practice; (vi) individual at high risk of developing the disease. The criteria for the 'public health benefit' argument were (i) substantial danger to public health; (ii) condition must have serious consequences if transmitted; (iii) effectiveness of the intervention; (iv) degree of invasiveness of the intervention; (v) whether individual receives an appreciable benefit not dependent on speculation about future behavior; (vi) the health benefit to society must outweigh the human rights cost to the individual.

The authors evaluated prophylactic mastectomy and cosmetic ear surgery against the best interests of the child criteria, immunization against the benefit to public health criteria and childhood male circumcision against both sets of criteria. They concluded that while immunization generally satisfied the best interests and public health justifications, infant circumcision satisfied neither. Such an intervention was thus impermissible because it was performed on a minor without the possibility of receiving his consent; the human rights cost to

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the individual exceeded the proven public health benefit; and the diseases from which circumcision might provide some level of protection could be avoided through appropriate behavioral choices or otherwise managed nonsurgically.

Individuals have a substantial interest in bodily integrity (including genital integrity) and autonomy (including what medical treatments to adopt). In contrast with quarantine, the harms of circumcision (as felt by those who do regard themselves as harmed by the procedure) are permanent and irreversible—loss of a valued external bodily structure that cannot be recovered, taken without consent (Watson, 2014; Earp and Darby, 2017; Hammond and Carmack, 2017). Moreover, childhood male circumcision is inequitable in that it targets only male infants and boys (leaving females, intersex children and adults alone), thus forcing them to bear the whole cost of whatever public health benefit is being pursued. It follows that the case for the permissibility of childhood male circumcision as a public health initiative must be far stronger than the case for the permissibility of either vaccination or quarantine—a case we have argued that McMath has not succeeded in establishing.

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Notes

1. Another problem with the current approach is that it ignores ongoing concerns about the reproducibility of basic findings across the biomedical and social sciences (Ioannidis, 2005, 2017; Collins and Tabak, 2014; Begley and Ioannidis, 2015; Earp and Trafimow, 2015), often taking contentious claims for granted while ignoring or downplaying concerns about researcher bias, politicking in peer review and agenda-driven science in this and other areas (Easterbrook *et al.*, 1991; Van Howe, 1999, 2017; Smith, 2010; Shaw, 2014; Earp and Darby, 2015; Ploug and Holm, 2015; Nissen *et al.*, 2016; Earp, 2016a, b; Bero, 2017).
2. Assuming that is what the average, all-things-considered non-circumcised male is. However, it is not clear that we are justified in assuming this, especially given the higher rates of male-to-female transmission of HIV among circumcised men in the fourth RCT (Wawer *et al.*, 2009).
3. It is conceivable that this analysis would differ if the foreskin had no functions or positive properties; if it were a uniquely vulnerable vector for diseases with serious public health ramifications (that could not be addressed in less ethically problematic ways); if the surgery could not be delayed without increasing the absolute risk of serious harms to an ethically decisive degree (see Earp and Darby, 2017); or if it were sound as a matter of general principle to construe healthy tissues as dangerous simply by virtue of being left on the body. Since none of these propositions hold, however, the prospect of non-consensual childhood circumcision is difficult to justify from a public health perspective.

References

American Academy of Pediatrics (AAP). (2012). Male Circumcision (Technical Report). *Pediatrics*, **130**, e756–e785.

Abdulcadir, J., Ahmadu, F. S., Essen, B., Gruenbaum, E., Johnsdotter, S., Johnson, M. C., Johnson-Agbakwu, C., Kratz, C., Sulkin, C. L., McKinley, M., Wairimu, N., Rogers, J., Shell-Duncan, B., and Shweder, R. A. (2012). Seven Things to Know About Female Genital Surgeries in Africa. *The Hastings Center Report*, **42**, 19–27.

Abu-Sahlieh, S. A. A. (1994). To Mutilate in the Name of Jehovah or Allah: Legitimization of Male and Female Circumcision. *Medicine and Law* **13**, 575–622.

Adler, P. (2016). Resolving Circumcision Controversies: The Burden of Proof and the Benefit of the Doubt [WWW Document]. *Circumcision and the Law* Blog, available from: https://circumcisionandthelawblog.wordpress.com/2016/03/30/resolving-circumcision-controversies-the-burden-of-proof-and-the-benefit-of-the-doubt/ [accessed 11 November 2017].

Aggleton, P. (2007). "Just a Snip"? A Social History of Male Circumcision. *Reproductive Health Matters*, **15**, 15–21.

Alani, A., Kelleher, M., Hemmings, K., Saunders, M., Hunter, M., Barclay, S., Ashley, M., Djemal, S., Bishop, K., Darbar, U., Briggs, P., and Fearne, J. (2015). Balancing the Risks and Benefits Associated with Cosmetic Dentistry—A Joint Statement by UK Specialist Dental Societies. *British Dental Journal*, **218**, 543–548.

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Alkhenizan, A. and Elabd, K. (2009). Non-therapeutic Infant Male Circumcision. Evidence, Ethics, and International Law Perspectives. *Saudi Medical Journal*, **37**, 941–947.

Androus, Z. T. (2004). The United States, FGM, and Global Rights to Bodily Integrity. *Presented at the The Rothermere American Institute Conference*. The United States and Global Human Rights. University of Oxford, pp. 1–7.

Auvert, B., Taljaard, D., Lagarde, E., Sobngwi-Tambekou, J., Sitta, R., Puren, A., and Deeks, S. (2005). Randomized, Controlled Intervention Trial of Male Circumcision for Reduction of HIV Infection Risk: The ANRS 1265 Trial. *PLoS Medicine*, **2**, e298.

Bailey, R. C., Moses, S., Parker, C. B., Agot, K., Maclean, I., Krieger, J. N., Williams, C. F., Campbell, R. T., and Ndinya-Achola, J. O. (2007). Male Circumcision for HIV Prevention in Young Men in Kisumu, Kenya: A Randomised Controlled Trial. *The Lancet*, **369**, 643–656.

Bailey, R. C., Nyaboke, I., Otieno, F. O., and Dezzutti, C. S. (2017). What Device Would be Best for Early Infant Male Circumcision in East and Southern Africa? Provider Experiences and Opinions with Three Different Devices in Kenya. *PLoS One*, **12**, e0171445.

Ball, P. J. (2006). A Survey of Subjective Foreskin Sensation in 600 Intact Men. In Denniston, G. C., Grassivaro, Gallo, P., Hodges, F. M., Milos, M. F. and Vivani, F. (eds), *Bodily Integrity and the Politics of Circumcision*, New York: Springer, pp. 177–188.

Begley, C. G. and Ioannidis, J. P. A. (2015). Reproducibility in Science. *Circulation Research*, **116**, 116–126.

Bell, K. (2015). HIV Prevention: Making Male Circumcision the "Right" Tool for the Job. *Global Public Health*, **10**, 552–572.

Benatar, M. and Benatar, D. (2003). Between Prophylaxis and Child Abuse: The Ethics of Neonatal Male Circumcision. *The American Journal of Bioethics*, **3**, 35–48.

Ben-Yami, H. (2013). Circumcision: What Should be Done? *Journal of Medical Ethics*, **39**, 459–462.

Bero, L. (2017). Addressing Bias and Conflict of Interest Among Biomedical Researchers. *JAMA*, **317**, 1723–1724.

Bill & Melinda Gates Foundation. (2017). *HIV Strategy Overview* [WWW Document]. Bill & Melinda Gates Foundation, available from: http://www.gatesfoundation.org/What-We-Do/Global-Health/HIV [accessed 29 June 2017].

British Medical Association (BMA). (2004). The Law and Ethics of Male Circumcision: Guidance for Doctors. *Journal of Medical Ethics*, **30**, 259–263.

Bossio, J. A. and Pukall, C. F. (2017). Attitude Toward One's Circumcision Status is More Important Than Actual Circumcision Status for Men's Body Image and Sexual Functioning. *Archives of Sexual Behavior*, in press.

Bossio, J. A., Pukall, C. F., and Steele, S. S. (2016). Examining Penile Sensitivity in Neonatally Circumcised and Intact Men Using Quantitative Sensory Testing. *The Journal of Urology*, **195**, 1848–1853.

Bossio, J. A., Pukall, C. F., and Steele, S. S. (2015). Response to: The Literature Supports Policies Promoting Neonatal Male Circumcision in N. America. *The Journal of Sexual Medicine*, **12**, 1306–1307.

Bossio, J. A., Pukall, C. F., and Steele, S. S. (2014). A Review of the Current State of the Male Circumcision Literature. *The Journal of Sexual Medicine*, **11**, 2847–2864.

Boyle, G. J., Svoboda, J., Price, C., and Turner, J. N. (2000). Circumcision of Healthy Boys: Criminal Assault? *Journal of Law and Medicine*, **7**, 301–310.

Bundick, S. (2009). Promoting Infant Male Circumision to Reduce Transmission of HIV: A Flawed Policy for the US [WWW Document]. *Health and Human Rights Journal* Blog, available from: https://www.hhrjournal.org/2009/08/promoting-infant-male-circumcision-to-reduce-transmission-of-hiv-a-flawed-policy-for-the-us/ [accessed 24 December 2016].

Carmack, A., Notini, L., and Earp, B. D. (2016). Should Surgery for Hypospadias be Performed Before an Age of Consent? *Journal of Sex Research*, **53**, 1047–1058.

Cartwright, N. (2011). A Philosopher's View of the Long Road from RCTs to Effectiveness. *The Lancet*, **377**, 1400–1401.

Cartwright, N. (2007). Are RCTs the Gold Standard? *BioSocieties*, **2**, 11–20.

Cartwright, N. and Joyce, K. (2017). Meeting Our Standards for Educational Justice: Doing Our Best with the Evidence. Department of Philosophy, University of San Diego. Unpublished Manuscript, 1–19.

Centers for Disease Control (CDC). (2014). *Draft CDC Recommendations for Providers Counseling Male Patients and Parents Regarding Male Circumcision*

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and the Prevention of HIV Infection, STIs, and Other Health Outcomes. US Center for Disease Control and Prevention.

Cetron, M. and Landwirth, J. (2005). Public Health and Ethical Considerations in Planning for Quarantine. *The Yale Journal of Biology and Medicine*, **78**, 325–3330.

Coghlan, A. (2009). Bill Gates Helps Fund Mass Circumcision Programme [WWW Document]. *New Science*, available from: https://www.newscientist.com/article/dn17312-bill-gates-helps-fund-mass-circumcision-programme/ [accessed 29 June 2017].

Cold, C. J. and Taylor, J. R. (1999). The Prepuce. *BJU International*, **83**, 34–44.

Collier, R. (2012). Ugly, Messy and Nasty Debate Surrounds Circumcision. *Canadian Medical Association Journal*, **184**, E25–E26.

Collins, F. S. and Tabak, L. A. (2014). NIH Plans to Enhance Reproducibility. *Nature*, **505**, 612–613.

Canadian Paediatric Society (CPS). (2015). Newborn Male Circumcision. *Paediatrics & Child Health*, **20**, 311–320.

Crosby, R. and Charnigo, R. J. (2013). A Comparison of Condom Use Perceptions and Behaviours between Circumcised and Intact Men Attending Sexually Transmitted Disease Clinics in the United States. *International Journal of STD and AIDS* **24**, 175–178.

Cruz, R., Glick, L. B., and Travis, J. W. (2003). Circumcision as Human-Rights Violation: Assessing Benatar and Benatar. *The American Journal of Bioethics*, **3**, 19–20.

Darby, R. (2016). Targeting Patients Who Cannot Object? Re-examining the Case for Non-Therapeutic Infant Circumcision. *SAGE Open*, **6**, 215824401664921.

Darby, R. (2015a). Risks, Benefits, Complications and Harms: Neglected Factors in the Current Debate on Non-therapeutic Circumcision. *Kennedy Institute of Ethics Journal*, **25**, 1–34.

Darby, R. (2015b). Syphilis 1855 and HIV-AIDS 2007: Historical Reflections on the Tendency to Blame Human Anatomy for the Action of Micro-organisms. *Global Public Health*, **10**, 573–588.

Darby, R. (2005). *A Surgical Temptation: The Demonization of the Foreskin and the Rise of Circumcision in Britain*. Chicago: University of Chicago Press.

Darby, R. and Cox, L. (2009). Objections of a Sentimental Character: The Subjective Dimensions of Foreskin Loss. *Matatu Journal of African Cultural Sociology*, **37**, 145–168.

Darby, R. and Van Howe, R. S. (2011). Not a Surgical Vaccine: There is No Case for Boosting Infant Male Circumcision to Combat Heterosexual Transmission of HIV in Australia. *Australian and New Zealand Journal of Public Health* **35**, 459–465.

Davis, D. S. (2001). Male and Female Genital Alteration: A Collision Course with the Law. *Health Matrix*, **11**, 487–570.

Davis, D. S. (2013). Ancient Rites and New Laws: How Should We Regulate Religious Circumcision of Minors? *Journal of Medical Ethics*, **39**, 456–458.

de Camargo, K. R., de Oliveira Mendonça, A. L., Perrey, C., and Giami, A. (2015a). Making the Circumcision Controversy Controversial: Going Meta and Taking Aim at the Messenger(s): Reply to Wamai *et al*. *Global Public Health*, **10**, 667–671.

de Camargo, K. R., de Oliveira Mendonça, A. L., Perrey, C., and Giami, A. (2013). Male Circumcision and HIV: A Controversy Study on "Facts" and "Values". *Global Public Health*, **8**, 769–783.

de Camargo, K. R., Mendonça, A. L. O., Perrey, C., and Giami, A. (2015b). Letter to the Editor: A Response to Wamai *et al*. *Global Public Health*, **10**, 1252–1252.

Dekkers, W. (2009). Routine (Non-Religious) Neonatal Circumcision and Bodily Integrity: A Transatlantic Dialogue. *Kennedy Institute of Ethics Journal*, **19**, 125–146.

Dekkers, W., Hoffer, C., and Wils, J.-P. (2005). Bodily Integrity and Male and Female Circumcision. *Medicine, Health Care, and Philosophy*, **8**, 179–191.

DeLaet, D. L. (2012). Genital Autonomy, Children's Rights, and Competing Rights Claims in International Human Rights Law. *International Journal of Child Rights*, **20**, 554–583.

DeLaet, D. L. (2009). Framing Male Circumcision as a Human Rights Issue? Contributions to the Debate Over the Universality of Human Rights. *Journal of Human Rights* **8**, 405–426.

Denniston, G. C., Hodges, F. M., and Milos, M. F. (eds). (2007). *Male and Female Circumcision: Medical, Legal, and Ethical Considerations in Pediatric Practice*. New York: Springer Science & Business Media.

Denniston, G. C. and Milos, M. F. (eds). (1997). *Sexual Mutilations*. Boston, MA: Springer.

Douglas, H. (2009). *Science, Policy, and the Value-free Ideal*. Pittsburgh: University of Pittsburgh Press.

Drazen, J. M., Kanapathipillai, R., Campion, E. W., Rubin, E. J., Hammer, S. M., Morrissey, S., and Baden, L. R. (2014). Ebola and Quarantine. *The New England Journal of Medicine*, **371**, 2029–2030.

Downloaded from https://academic.oup.com/phe/advance-article-abstract/doi/10.1093/phe/phx024/4764011
by guest
on 20 December 2017

Dreger, A. (1999). *Intersex in the Age of Ethics*. Hagerstown: University Publishing Group.

Dushoff, J., Patocs, A., Shi, C.-F., and Colizza, V. (2011). Modeling the Population-Level Effects of Male Circumcision as an HIV-Preventive Measure: A Gendered Perspective. *PLoS One*, **6**, e28608.

Earp, B. D. (2017a). Does Female Genital Mutilation Have Health Benefits? The Problem With Medicalizing Morality [WWW Document]. *Quillette Magazine,* republished at *Practical Ethics* Blog, University of Oxford available from http://blog.practicalethics.ox.ac.uk/2017/08/does-female-genital-mutilation-have-health-benefits-the--problem-with-medicalizing-morality/ [accessed 11 November 2017].

Earp, B. D. (2017b). Gender, Genital Alteration, and Beliefs about Bodily Harm. *The Journal of Sexual Medicine (23rd Congress of the World Association for Sexual Health)*, **14**, e225.

Earp, B. D. (2017c). The Right To Bodily Integrity and the Concept of Sexual Harm. *The Journal of Sexual Medicine (23rd Congress of the World Association for Sexual Health)*, **14**, e239.

Earp, B. D. (2016a). The Unbearable Asymmetry of Bullshit. *Health Watch,* Spring, 4–5.

Earp, B. D. (2016b). Mental Shortcuts. *The Hastings Center Report*, **46**, inside front cover.

Earp, B. D. (2016c). Between Moral Relativism and Moral Hypocrisy: Reframing the Debate on "FGM". *Kennedy Institute of Ethics Journal*, **26**, 105–144.

Earp, B. D. (2016d). Infant Circumcision and Adult Penile Sensitivity: Implications for Sexual Experience. *Trends in Urology Men's Health*, **7**, 17–21.

Earp, B. D. (2015a). Addressing Polarisation in Science. *Journal of Medical Ethics*, **41**, 782–784.

Earp, B. D. (2015b). Do the Benefits of Male Circumcision Outweigh the Risks? A Critique of the Proposed CDC Guidelines. *Frontiers in Pediatrics*, **3**, 1–6.

Earp, B. D. (2015c). The Need to Control for Socially Desirable Responding in Studies on the Sexual Effects of Male Circumcision. *PLoS One*, **10**, 1–12.

Earp, B. D. (2015d). Sex and Circumcision. *The American Journal of Bioethics*, **15**, 43–45.

Earp, B. D. (2015e). Female Genital Mutilation and Male Circumcision: Toward an Autonomy-Based Ethical Framework. *Medicolegal and Bioethics*, **5**, 89–104.

Earp, B. D. (2013). The Ethics of Infant Male Circumcision. *Journal of Medical Ethics*, **39**, 418–420.

Earp, B. D. (2012). The AAP Report on Circumcision: Bad Science + Bad Ethics = Bad Medicine [WWW Document]. *Practical Ethics* Blog. University of Oxford, available from http://blog.practicalethics.ox.ac.uk/2012/08/the-aap-report-on-circumcision-bad-science-bad-ethics-bad-medicine/ [accessed 11 November 2017].

Earp, B. D. and Darby, R. (2017). Circumcision, Sexual Experience, and Harm. *University of Pennsylvania Journal of International Law*, **37** (online), 1–57.

Earp, B. D. and Darby, R. (2015). Does Science Support Infant Circumcision? A Skeptical Reply to Brian Morris. *The Skeptic*, **25**, 23–30.

Earp, B. D., Hendry, J., and Thomson, M. (2017a). Reason and Paradox in Medical and Family Law: Shaping Children's Bodies. *Medical Law Review*, in press.

Earp, B. D., Sardi, L. M., and Jellison, W. A. (2017b). False Beliefs Predict Increased Circumcision Satisfaction in a Sample of US American Men. *Culture, Health and Sexuality*, in press.

Earp, B. D. and Shaw, D. M. (2017). Cultural Bias in American Medicine: The Case of Infant Male Circumcision. *Journal of Pediatric Ethics*, **1**, 8–26.

Earp, B. D. and Steinfeld, D. R. (2017). *Gender and Genital Cutting: A New Paradigm.* In T. G. Barbat (Ed.), Gifted Women, Fragile Men. Euromind Monographs – 2, Brussels: ALDE Group-EU Parliament.

Earp, B. D. and Trafimow, D. (2015). Replication, Falsification, and the Crisis of Confidence in Social Psychology. *Frontiers in Psychology*, **6**, 1–11.

Easterbrook, P. J., Gopalan, R., Berlin, J. A., and Matthews, D. R. (1991). Publication Bias in Clinical Research. *The Lancet*, **337**, 867–872.

Feldblum, P. J., Okech, J., Ochieng, R., Hart, C., Kiyuka, G., Lai, J. J., Veena, V., and Hills, R. K. (2015). Longer-term Follow-up of Kenyan Men Circumcised Using the ShangRing Device. *PLoS One*, **10**, e0137510.

Foddy, B. (2013). Medical, Religious and Social Reasons for and Against an Ancient Rite. *Journal of Medical Ethics*, **39**, 415–415.

Fox, M. and Thomson, M. (2010). HIV/AIDS and Circumcision: Lost in Translation. *Journal of Medical Ethics*, **36**, 798–801.

Fox, M. and Thomson, M. (2005). Short Changed? The Law and Ethics of Male Circumcision. *International Journal of Children's Rights*. **13**, 161–181.

Frisch, M. (2016). Higher Frequency of Unprotected Insertive Anal Sex Among Young Black MSM Who Are Circumcised. *AIDS and Behavior*, **20**, 2543–2544.

Downloaded from https://academic.oup.com/phe/advance-article-abstract/doi/10.1093/phe/phx024/4764011
by guest
on 20 December 2017

Frisch, M., Aigrain, Y., Barauskas, V., Bjarnason, R., Boddy, S.-A., Czauderna, P., de Gier, R. P. E., de Jong, T. P. V. M., Fasching, G., Fetter, W., Gahr, M., Graugaard, C., Greisen, G., Gunnarsdottir, A., Hartmann, W., Havranek, P., Hitchcock, R., Huddart, S., Janson, S., Jaszczak, P., Kupferschmid, C., Lahdes-Vasama, T., Lindahl, H., MacDonald, N., Markestad, T., Märtson, M., Nordhov, S. M., Pälve, H., Petersons, A., Quinn, F., Qvist, N., Rosmundsson, T., Saxen, H., Söder, O., Stehr, M., von Loewenich, V. C. H., Wallander, J., and Wijnen, R. (2013). Cultural bias in the AAP's 2012 Technical Report and Policy Statement on Male Circumcision. *Pediatrics*, **131**, 796–800.

Frisch, M. and Earp, B. D. (2016). Circumcision of Male Infants and Children as a Public Health Measure in Developed Countries: A Critical Assessment of Recent Evidence. *Global Public Health*, in press.

Frisch, M. and Simonsen, J. (2016). Cultural Background, Non-Therapeutic Circumcision and the Risk of Meatal Stenosis and Other Urethral Stricture Disease: Two Nationwide Register-Based Cohort Studies in Denmark 1977–2013. *The Surgeon*, in press.

Frisch, M. and Simonsen, J. (2015a). Ritual Circumcision and Risk of Autism Spectrum Disorder in 0- to 9-year-old Boys: National Cohort Study in Denmark. *Journal of the Royal Society of Medicine*, **108**, 266–279.

Frisch, M. and Simonsen, J. (2015b). Circumcision-autism Link Needs Thorough Evaluation: Response to Morris and Wiswell. *Journal of the Royal Society of Medicine*, **108**, 297–298.

Gage, A. J. and Van Rossem, R. (2006). Attitudes Toward the Discontinuation of Female Genital Cutting Among Men and Women in Guinea. *International Journal of Gynecological Obstetrical*, **92**, 92–96.

Garenne, M., Giami, A., and Perrey, C. (2013). Male Circumcision and HIV Control in Africa: Questioning Scientific Evidence and the Decision-Making Process. In Giles-Vernik, T. and Webb, J. (eds), *Global Health in Africa: Historical Perspectives on Disease Control*. Athens: Ohio University Press, pp. 185–210.

Giami, A. and Perrey, C. (2012). Transformations in the Medicalization of Sex: HIV Prevention between Discipline and Biopolitics. *Journal of Sex Research*, **49**, 353–361.

Giami, A., Perrey, C., de Oliveira Mendonça, A. L., and de Camargo, K. R. (2015). Hybrid Forum or Network? The Social and Political Construction of an International "Technical Consultation": Male Circumcision and HIV Prevention. *Global Public Health*, **10**, 589–606.

Glick, L. B. (2005). *Marked in Your Flesh: Circumcision from Ancient Judea to Modern America*. Oxford: Oxford University Press.

Gollaher, D. L. (2000). *Circumcision: A History of the World's Most Controversial Surgery*. New York: Basic Books.

Gollaher, D. L. (1994). From Ritual to Science: The Medical Transformation of Circumcision in America. *Journal of Social History*, **28**, 5–36.

Gonsalves, G., and Staley, P. (2014). Panic, Paranoia, and Public Health—The AIDS Epidemic's Lessons for Ebola. *The New England Journal of Medicine*, **371**, 2348–2349.

Gray, R. H., Kigozi, G., Serwadda, D., Makumbi, F., Watya, S., Nalugoda, F., Kiwanuka, N., Moulton, L. H., Chaudhary, M. A., Chen, M. Z., Sewankambo, N. K., Wabwire-Mangen, F., Bacon, M. C., Williams, C. F., Opendi, P., Reynolds, S. J., Laeyendecker, O., Quinn, T. C., and Wawer, M. J. (2007). Male Circumcision for HIV Prevention in Men in Rakai, Uganda: A Randomised Trial. *The Lancet*, **369**, 657–666.

Grund, J. M., and Hennink, M. M. (2012). A Qualitative Study of Sexual Behavior Change and Risk Compensation Following Adult Male Circumcision in Urban Swaziland. *AIDS Care*, **24**, 245–251.

Hammond, T. (1999). A Preliminary Poll of Men Circumcised in Infancy or Childhood. *BJU International*, **83**, 85–92.

Hammond, T. and Carmack, A. (2017). Long-term Adverse Outcomes from Neonatal Circumcision Reported in a Survey of 1, 008 Men: An Overview of Health and Human Rights Implications. *International Journal of Human Rights*, **21**, 189–218.

Hatzold, K., Mavhu, W., Jasi, P., Chatora, K., Cowan, F. M., Taruberekera, N., Mugurungi, O., Ahanda, K., Njeuhmeli, E., and Tang, J. W. (2014). Barriers and Motivators to Voluntary Medical Male Circumcision Uptake Among Different Age Groups of Men in Zimbabwe: Results from a Mixed Methods Study. *PLoS One*, **9**, e85051.

Hladik, F. and Hope, T. J. (2009). HIV Infection of the Genital Mucosa in Women. *Current HIV/AIDS Reports*, **6**, 20–28.

Hodges, F. (1997). A Short History of the Institutionalization of Involuntary Sexual Mutilation in the United States. In Denniston, G. C. and Milos, M. F. (eds), *Sexual Mutilations*. New York: Springer, pp. 17–40.

Downloaded from https://academic.oup.com/phe/advance-article-abstract/doi/10.1093/phe/phx024/4764011
by guest
on 20 December 2017

Hodges, F. M., Svoboda, J. S., and Van Howe, R. S. (2002). Prophylactic Interventions on Children: Balancing Human Rights With Public Health. *Journal of Medical Ethics*, **28**, 10–16.

Hodson, N. and Bewley, S. (2017). Pursuing Ethical Coherence in the Prevention of Vertical Transmission of HIV: Justice and Injustice in Option B + . *Journal of Virus Eradication*, **3**, 163–166.

Holm, S. (2004). Irreversible Bodily Interventions in Children. *Journal of Medical Ethics*, **30**, 237–237.

Ioannidis, J. P. A. (2017). Hijacked Evidence-Based Medicine: Stay the Course and Throw the Pirates Overboard. *Journal of Clinical Epidemiology*, **84**, 11–13.

Ioannidis, J. P. A. (2005). Why Most Published Research Findings Are False. *PLoS Medicine*, **2**, e124.

Jacobs, A. J. and Arora, K. S. (2015). Ritual Male Infant Circumcision and Human Rights. *The American Journal of Bioethics*, **15**, 30–39.

Jonsen, A. R. (1978). Do No Harm. *Annals of Internal Medicine*, **88**, 827–832.

Jung, J., Kim, B., Kim, H. B., and Cristian, P.-E. (2016). Long-term Effects of Male Circumcision on Risky Sexual Behaviors and STD Infections: Evidence from Malawian Schools. KDI School Working Paper Series. Working Paper 16-08, pp. 1–43. Available from https://ssrn.com/abstract=2833153 [accessed 11 November 2017].

Karkazis, K. (2008). *Fixing Sex: Intersex, Medical Authority, and Lived Experience*. Durham: Duke University Press.

Katisi, M. and Daniel, M. (2015). Safe Male Circumcision in Botswana: Tension between Traditional Practices and Biomedical Marketing. *Global Public Health*, **10**, 739–756.

Kibira, S. P. S., Atuyambe, L. M., Sandøy, I. F., Makumbi, F. E., and Daniel, M. (2017). "Now That You Are Circumcised, You Cannot Have First Sex With Your Wife": Post Circumcision Sexual Behaviours and Beliefs Among Men in Wakiso District, Uganda. *Journal of the International AIDS Society*, **20**, 21498.

Kigozi, G., Wawer, M., Ssettuba, A., Kagaayi, J., Nalugoda, F., Watya, S., Mangen, F. W., Kiwanuka, N., Bacon, M. C., Lutalo, T., Serwadda, D., and Gray, R. H. (2009). Foreskin Surface Area and HIV Acquisition in Rakai, Uganda (Size Matters). *AIDS*, **23**, 2209–2213.

Kipnis, K. and Diamond, M. (1998). Pediatric Ethics and the Surgical Assignment of Sex. *The Journal of Clinical Ethics*, **9**, 398–410.

Kluge, E. -H. (1994). Dr. Kluge Responds. *Canadian Medical Association Journal*, **150**, 1542.

KNMG. (2010). *Nontherapeutic Circumcision of Male Minors*. The Royal Dutch Medical Association. KNMG, pp. 1–17. Available from http://www.circumstitions.com/Docs/KNMG-policy.pdf [accessed 11 November, 2017]

Kopelman, L. M. (2014). Make Her a Virgin Again: When Medical Disputes about Minors Are Cultural Clashes. *The Journal of Medicine and Philosophy*, **39**, 8–25.

Kopelman, L. M. (2007). The Best Interests Standard for Incompetent or Incapacitated Persons of All Ages. *Journal of Law, Medicine and Ethics*, **35**, 187–196.

Kopelman, L. M. and Kopelman, A. E. (2007). Using a New Analysis of the Best Interests Standard to Address Cultural Disputes: Whose Data, Which Values? *Theoretical Medicine and Bioethics*, **28**, 373–391.

Levesque, A. and Li, H. Z. (2014). The Relationship between Culture, Health Conceptions, and Health Practices: A Qualitative–Quantitative Approach. *Journal of Cross-Cultural Psychology*, **45**, 628–645.

Lightfoot-Klein, H., Chase, C., Hammond, T., and Goldman, R. (2000). Genital Surgery on Children Below the Age of Consent. In Szuchman, L. T. and Muscarella, F. (eds), *Psychological Perspectives on Human Sexuality*, New York: John Wiley & Sons, pp. 440–479.

Ludbrook, R. (1995). The Child's Right to Bodily Integrity. *Current Issues in Criminal Justice*, **7**, 123–132.

Lyons, B. (2013). Male Infant Circumcision as a "HIV Vaccine". *Public Health Ethics*, **6**, 90–103.

Maslen, H., Earp, B. D., Cohen Kadosh, R., and Savulescu, J. (2014). Brain Stimulation for Treatment and Enhancement in Children: An Ethical Analysis. *Frontiers in Human Neuroscience*, **8**, 1–5.

Mavhu, W., Hatzold, K., Laver, S. M., Sherman, J., Tengende, B. R., Mangenah, C., Langhaug, L. F., Hart, G., Cowan, F. M., and Thorne, C. (2012). Acceptability of Early Infant Male Circumcision as an HIV Prevention Intervention in Zimbabwe: A Qualitative Perspective. *PLoS One*, **7**, e32475.

Mazor, J. (2013). The Child's Interests and the Case for the Permissibility of Male Infant Circumcision. *Journal of Medical Ethics*, **39**, 421–428.

McMath, A. (2015a). Infant Male Circumcision and the Autonomy of the Child: Two Ethical Questions. *Journal of Medical Ethics*, **41**, 687–690.

McMath, A. (2015b). Autonomy and the Circumcision Wars [WWW Document]. *Journal of Medical Ethics*

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Blog. Available from: http://blogs.bmj.com/medical-ethics/2015/02/27/autonomy-and-the-circumcision-wars/ [accessed 29 June 2017].

Merkel, R. and Putzke, H. (2013). After Cologne: Male Circumcision and the Law. Parental Right, Religious Liberty or Criminal Assault? *Journal of Medical Ethics*, **39**, 444–449.

Miller, G. P. (2002). Circumcision: Cultural-Legal Analysis. *Virginia Journal of Social Policy and the Law*, **9**, 497–585.

Mollov, N. D., Lindauer, S. J., Best, A. M., Shroff, B., and Tufekci, E. (2010). Patient Attitudes Toward Retention and Perceptions of Treatment Success. *The Angle Orthodontist*, **80**, 656–661.

Nissen, S. B., Magidson, T., Gross, K., and Bergstrom, C. T. (2016). Publication Bias and the Canonization of False Facts. *eLife*, **5**, e21451.

Norton, A. T. (2013). Surveying Risk Subjects: Public Health Surveys as Instruments of Biomedicalization. *BioSocieties*, **8**, 265–288.

Norton, A. T. (2017). Foreskin and the Molecular Politics of Risk. *Social Studies of Science*, **47**, 655–680.

Ploug, T. and Holm, S. (2015). Conflict of Interest Disclosure and the Polarisation of Scientific Communities. *Journal of Medical Ethics*, **41**, 356–358.

Price, C. (1997). Male Circumcision: An Ethical and Legal Affront. *Bulletin of Medical Ethics*, **128**, 13–19.

RACP. (2010). *Circumcision of Infant Males*. The Royal Australasian College of Physicians, pp. 1–28. Available from https://www.racp.edu.au/docs/default-source/advocacy-library/circumcision-of-infant-males.pdf [accessed 11 November 2017].

Reis, E. (2009). *Bodies in Doubt: An American History of Intersex*. Baltimore, MD: Johns Hopkins University Press.

Richters, J. (2009). Bodies, Pleasure and Displeasure. *Culture, Health and Sexuality*, **11**, 225–236.

Richters, J., Gerofi, J., and Donovan, B. (1995). Why Do Condoms Break or Slip Off in Use? An Exploratory Study. *International Journal of STD and AIDS*, **6**, 11–18.

Riess, T. H., Achieng', M. M., Otieno, S., Ndinya-Achola, J. O., Bailey, R. C., and Myer, L. (2010). "When I Was Circumcised I Was Taught Certain Things": Risk Compensation and Protective Sexual Behavior Among Circumcised Men in Kisumu, Kenya. *PLoS One*, **5**, e12366.

Rudrum, S., Oliffe, J. L., and Benoit, C. (2017). Discourses of Masculinity, Femininity and Sexuality in Uganda's Stand Proud, Get Circumcised Campaign. *Culture, Health and Sexuality*, **19**, 225–239.

Savulescu, J. (2013). Male Circumcision and the Enhancement Debate: Harm Reduction, Not Prohibition. *Journal of Medical Ethics*, **39**, 416–417.

Schoen, E. J. (2006). Ignoring Evidence of Circumcision Benefits. *Pediatrics*, **118**, 385–387.

Scott, S. (1999). The Anatomy and Physiology of the Human Prepuce. In Denniston, G. C., Hodges, F. M. and Milos, M. F. (eds), *Male and Female Circumcision*. New York: Springer, pp. 9–18.

Shaw, D. M. (2014). Beyond Conflicts of Interest: Disclosing Medical Biases. *Journal of the American Medical Association*, **312**, 697–698.

Shi, C. F., Li, M., and Dushoff, J. (2017). Evidence that Promotion of Male Circumcision did not Lead to Sexual Risk Compensation in Prioritized Sub-Saharan Countries. *PLoS One*, **12**, e0175928.

Shweder, R. A. (2016). Equality Now in Genital Reshaping: Brian Earp's Search for Moral Consistency. *Kennedy Institute of Ethics Journal*, **26**, 145–154.

Shweder, R. A. (2013). The Goose and the Gander: The Genital Wars. *Glob Discourse*, **3**, 348–366.

Sidler, D., Earp, B. D., van Niekerk, A., Moodley, K., and Kling, S. (2017). Targeting Mothers and Selling Men What They Do Not Want: A Response to "Missed Opportunities for Circumcision of Boys." *South African Medical Journal*, **107**, 281.

Sidler, D., Smith, J., and Rode, H. (2008). Neonatal Circumcision Does Not Reduce HIV/AIDS Infection Rates. *SAMJ South African Medical Journal*, **98**, 762–766.

Silverman, E. K. (2004). Anthropology and Circumcision. *Annual Review of Anthropology*, **33**, 419–445.

Sircili, M. H., e Silva, F. A., Costa, E. M., Brito, V. N., Arnhold, I. J., Dénes, F. T., Inacio, M., and de Mendonca, B. B. (2010). Long-term Surgical Outcome of Masculinizing Genitoplasty in Large Cohort of Patients with Disorders of Sex Development. *The Journal of Urology*, **184**, 1122–1127.

Smith, R. (2010). Classical Peer Review: An Empty Gun. *Breast Cancer Research*, **12**, S13.

Sommerville, M. (2004). *The Ethical Canary: Science, Society, and the Human Spirit*. Montreal: McGill-Queen's University Press.

Sorrells, M. L., Snyder, J. L., Reiss, M. D., Eden, C., Milos, M. F., Wilcox, N., and Van Howe, R. S. (2007). Fine-touch Pressure Thresholds in the Adult Penis. *BJU International*, **99**, 864–869.

Svoboda, J. S. (2013). Promoting Genital Autonomy by Exploring Commonalities between Male, Female,

Downloaded from https://academic.oup.com/phe/advance-article-abstract/doi/10.1093/phe/phx024/4764011
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on 20 December 2017

Intersex, and Cosmetic Female Genital Cutting. *Global Discourse*, **3**, 237–255.

Svoboda, J. S., Adler, P. W., and Van Howe, R. S. (2016). Circumcision is Unethical and Unlawful. *Journal of Law, Medicine and Ethics*, **44**, 263–282.

Taube, J. M., Nichols, A. D., Bornman, L. S., Bornman, D. M., and Jackson, J. B. (2007). Langerhans Cell Density and High-Grade Vulvar Intraepithelial Neoplasia in Women With Human Immunodeficiency Virus Infection. *Journal of Cutaneous Pathology*, **34**, 565–570.

Taylor, J. R., Lockwood, A. P., and Taylor, A. J. (1996). The Prepuce: Specialized Mucosa of the Penis and Its Loss to Circumcision. *British Journal of Urology*, **77**, 291–295.

Tobian, A. A. and Gray, R. H. (2011). The Medical Benefits of Male Circumcision. *JAMA*, **306**, 1479–1480.

Ungar-Sargon, E. (2015). On the Impermissibility of Infant Male Circumcision: A Response to Mazor (2013). *Journal of Medical Ethics*, **41**, 186–190.

Van Howe, R. S. (2017). Expertise or Ideology? A Response to Morris *et al.* 2016, "Circumcision is a Primary Preventive Against HIV Infection: Critique of a Contrary Meta-Regression Analysis by Van Howe." *Global Public Health*, in press.

Van Howe, R. S. (2013). Infant Circumcision: The Last Stand for the Dead Dogma of Parental (Sovereignal) Rights. *Journal of Medical Ethics*, **39**, 475–481.

Van Howe, R. S. (1999). Peer-review Bias Regarding Circumcision in American Medical Publishing. In Denniston, G. C., Hodges, F. M., and Milos, M. F. (eds), *Male and Female Circumcision*. New York: Springer, pp. 357–378.

Van Howe, R. S. and Storms, M. R. (2011). How the Circumcision Solution in Africa Will Increase HIV Infections. *Journal of Public Health in Africa*, **2**, 11–15.

Waldeck, S. E. (2003). Using Male Circumcision to Understand Social Norms as Multipliers. *University of Cincinnati Law Review*, **72**, 455–526.

Wallerstein, E. J. (1985). Circumcision: The Uniquely American Medical Enigma. *The Urologic Clinics of North America*, **12**, 123–132.

Wallerstein, E. J. (1980). *Circumcision: An American Health Fallacy*. New York: Springer.

Watson, L. (2014). *Unspeakable Mutilations: Circumcised Men Speak Out*. Ashburton, NZ: Amazon Media.

Wawer, M. J., Makumbi, F., Kigozi, G., Serwadda, D., Watya, S., Nalugoda, F., Buwembo, D., Ssempijja, V., Kiwanuka, N., Moulton, L. H., Sewankambo, N. K., Reynolds, S. J., Quinn, T. C., Opendi, P., Iga, B., Ridzon, R., Laeyendecker, O., and Gray, R. H. (2009). Circumcision in HIV-Infected Men and Its Effect on HIV Transmission to Female Partners in Rakai, Uganda: A Randomised Controlled Trial. *The Lancet*, **374**, 229–237.

Werker, P. M., Terng, A. S., and Kon, M. (1998). The Prepuce Free Flap: Dissection Feasibility Study and Clinical Application of a Super-Thin New Flap. *Plastic and Reconstructive Surgery*, **102**, 1075–1082.

Wisdom, T. (2012). Advocating Genital Autonomy: Methods of Intactivism in the United States. *Research Journal for the Human Science*, **11,** available from: http://www.kon.org/urc/v11/wisdom.html.

Downloaded from https://academic.oup.com/phe/advance-article-abstract/doi/10.1093/phe/phx024/4764011
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